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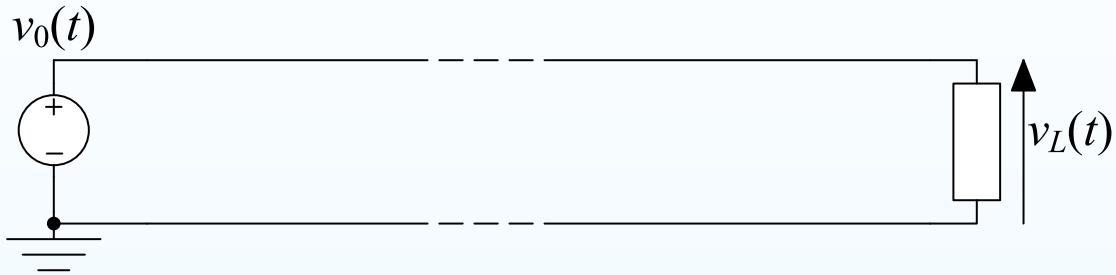

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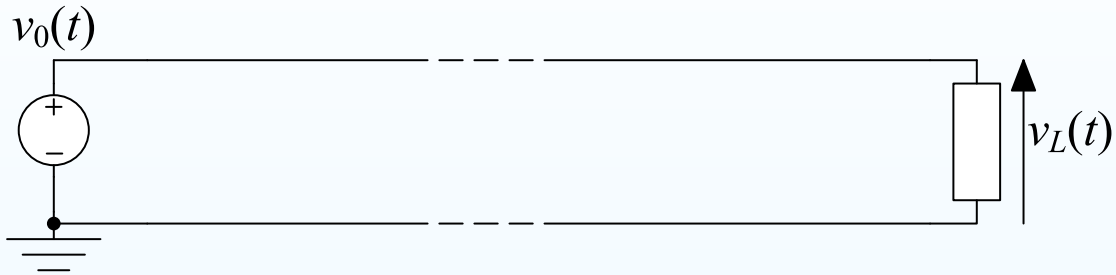

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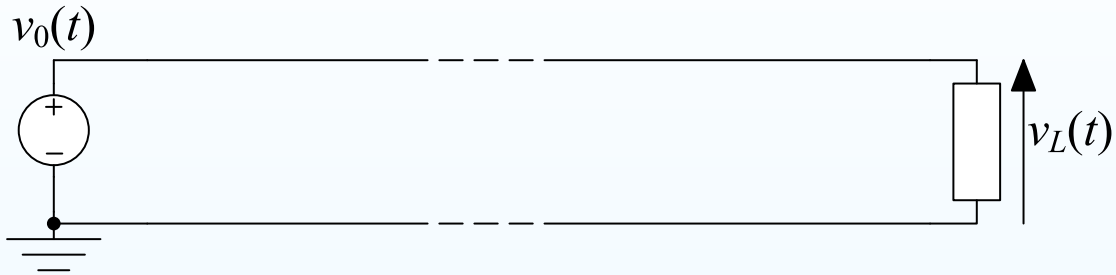

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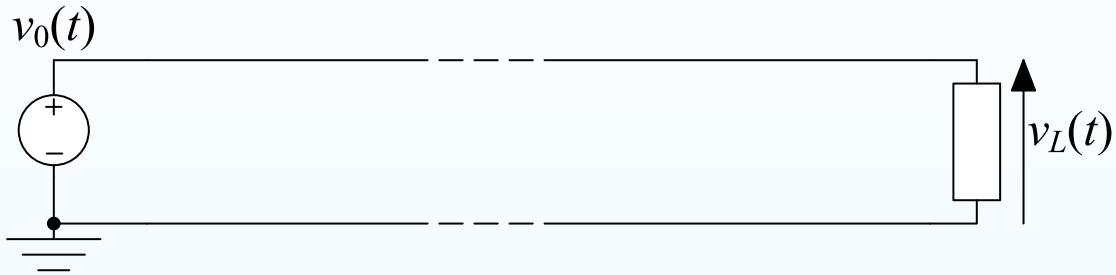

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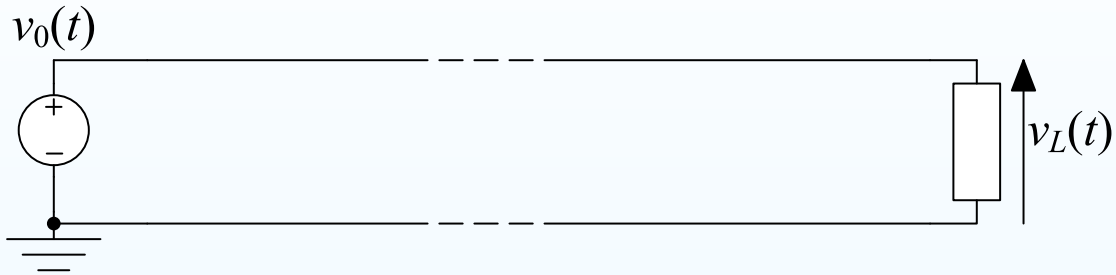

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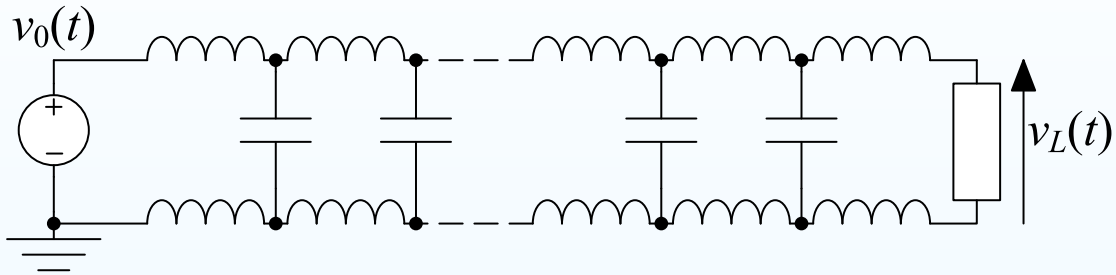

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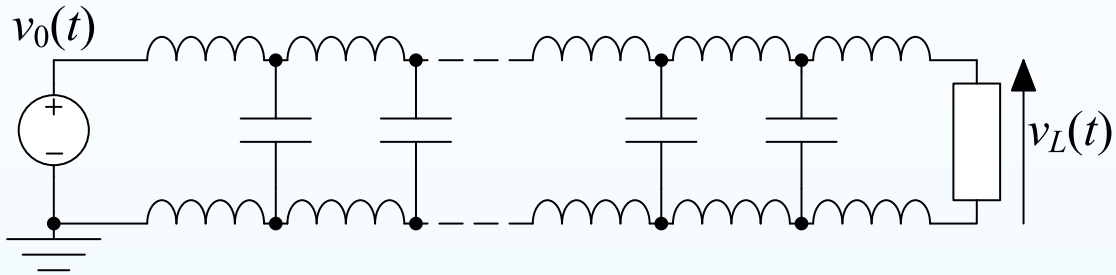

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Transmission Line Equations

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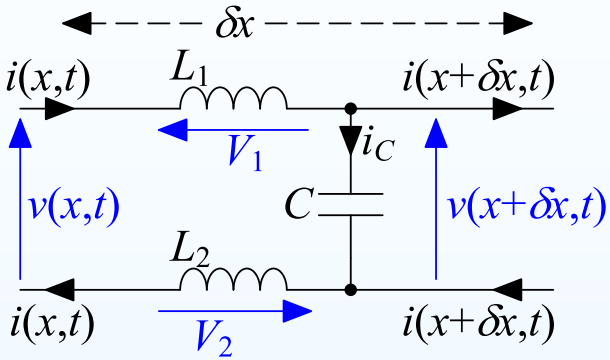
A short section of line δx long:

$v(x,t)$ and $i(x,t)$ depend on both position and time.



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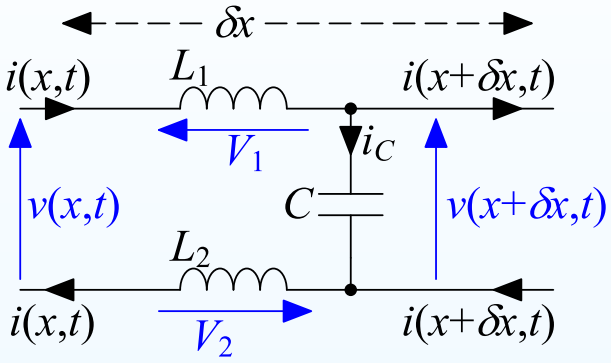

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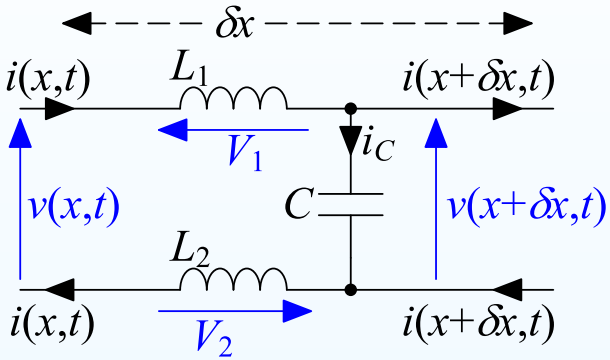

Basic Equations

KVL: $v(x,t) = V_2 + v(x+\delta x,t) + V_1$

KCL: $i(x,t) = i_C + i(x+\delta x,t)$

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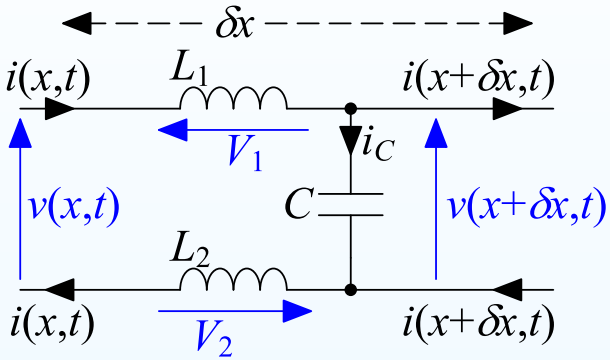

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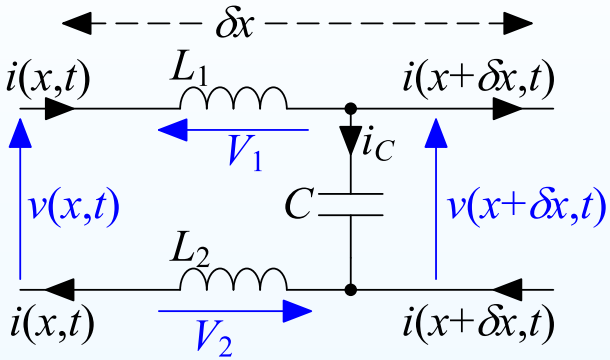

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$$(L_1 + L_2)\frac{\partial i}{\partial t} = V_1 + V_2 = v(x,t) - v(x+\delta x,t) = -\frac{\partial v}{\partial x}\delta x$$

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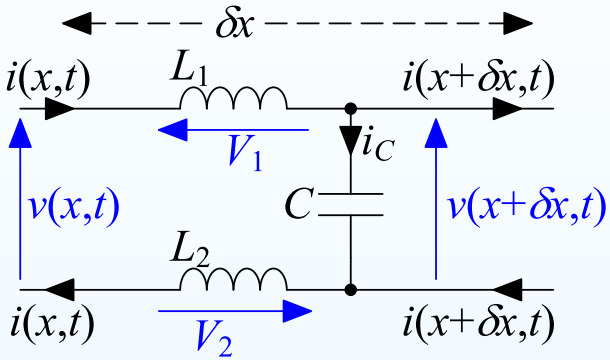

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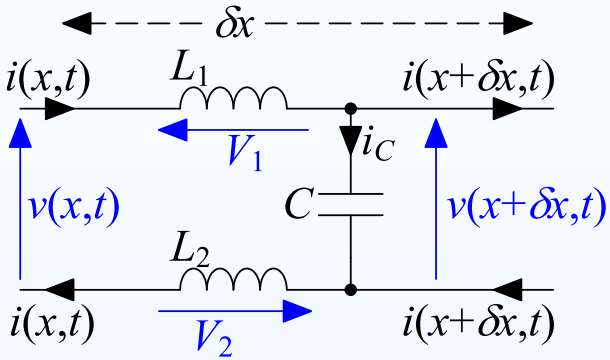

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where $C_0 = \frac{C}{\delta x}$ is the capacitance per unit length (Farads/m) and $L_0 = \frac{L_1+L_2}{\delta x}$ is the total inductance per unit length (Henries/m).

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Verify by substitution:

$$-\frac{\partial i}{\partial x} = -\left(\frac{-f'(t-\frac{x}{u}) - g'(t+\frac{x}{u})}{Z_0} \times \frac{1}{u}\right)$$

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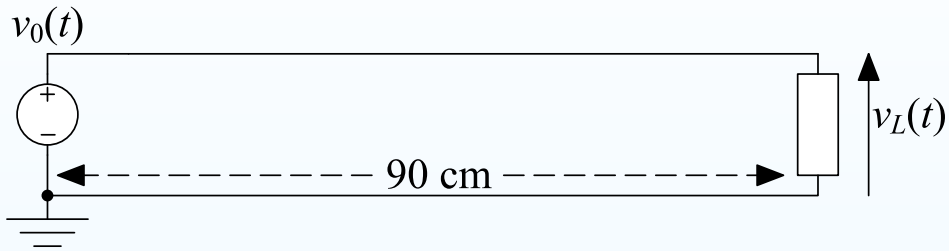
Suppose:

$u = 15\,\mathrm{cm/ns}$

and $g(t) \equiv 0$

$\Rightarrow v(x,t) = f\left(t - \frac{x}{u}\right)$



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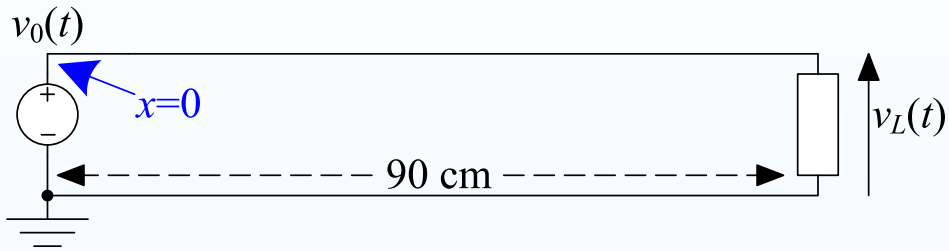
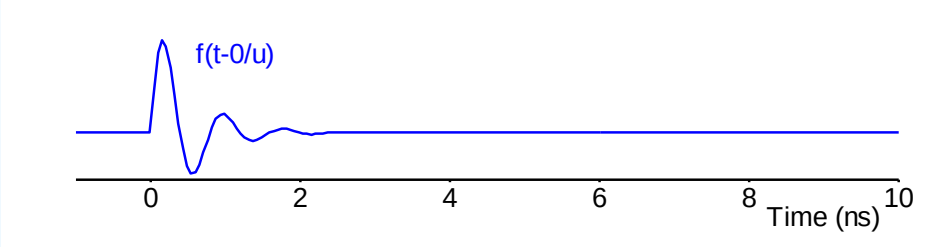
- At $x = 0$ cm [▲], $v_S(t) = f(t - \frac{0}{u})$





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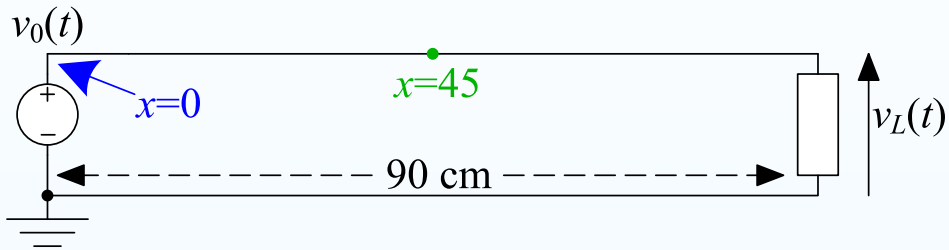
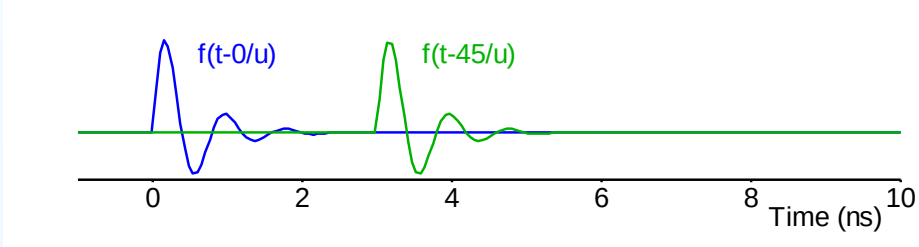
- At $x = 0$ cm [▲], $v_S(t) = f(t - \frac{0}{u})$
- At $x = 45$ cm [▲], $v(45,t) = f(t - \frac{45}{u})$





Forward Wave

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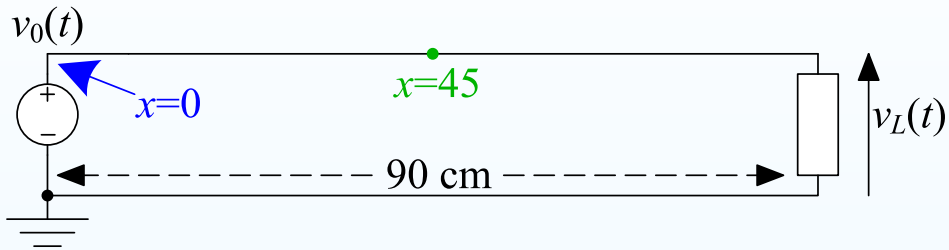
- Summary

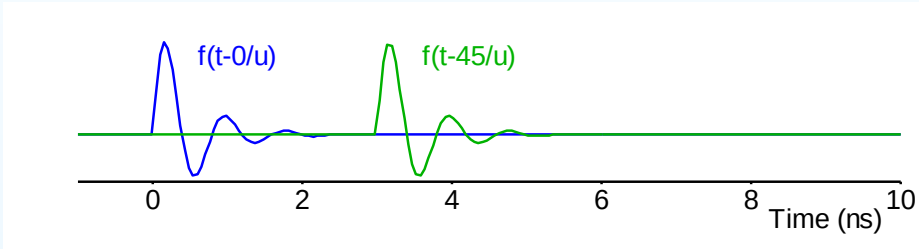
Suppose:

$u = 15\,\mathrm{cm/ns}$

and $g(t) \equiv 0$

$\Rightarrow v(x,t) = f\left(t - \frac{x}{u}\right)$





- At $x = 0$ cm [▲],
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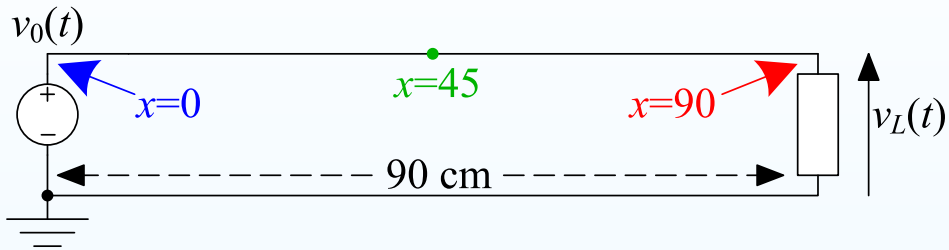

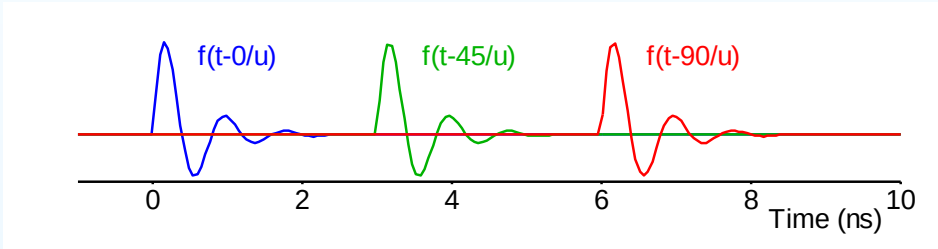

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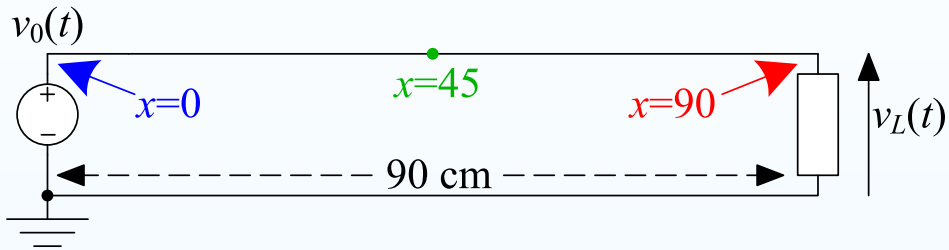

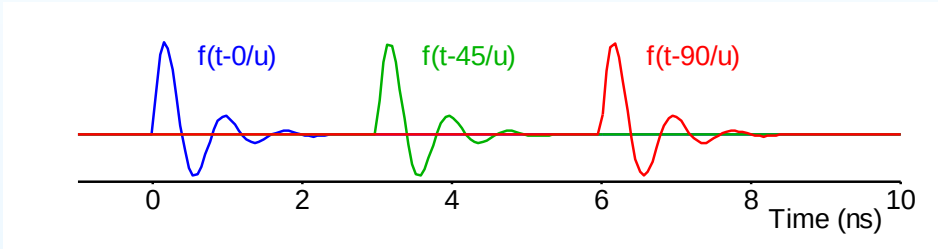

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Waveform at $x = 0$ completely determines the waveform everywhere else.

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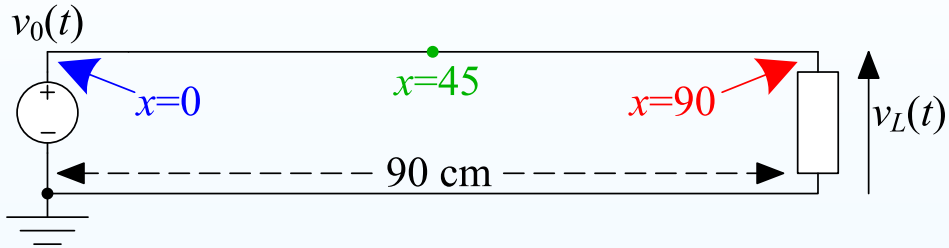

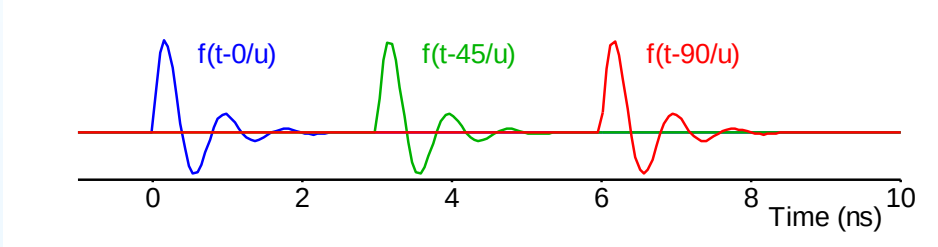

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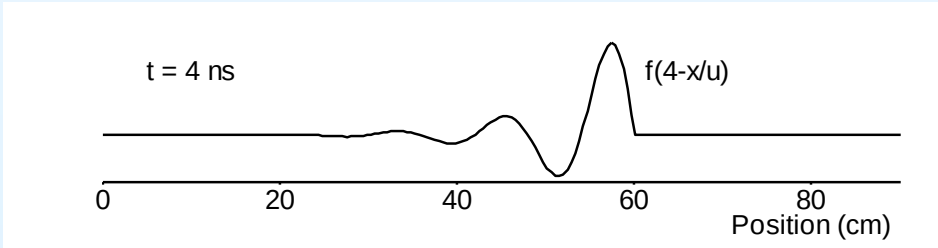
Snapshot at $t_0 = 4\,\mathrm{ns}$:
the waveform has just arrived at the point $x = ut_0 = 60$ cm.



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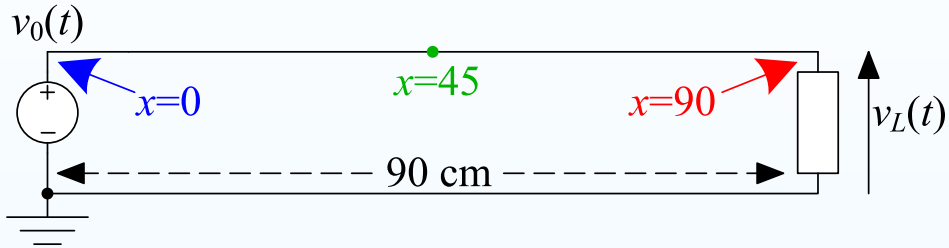

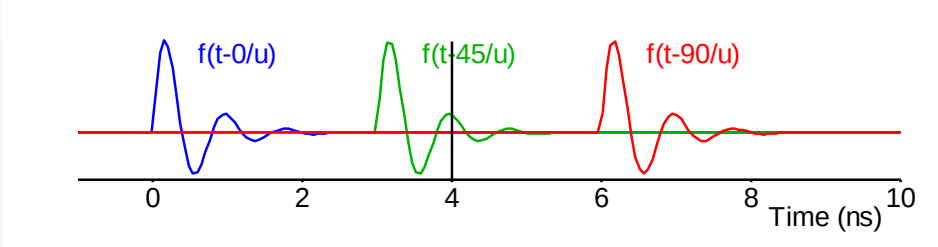

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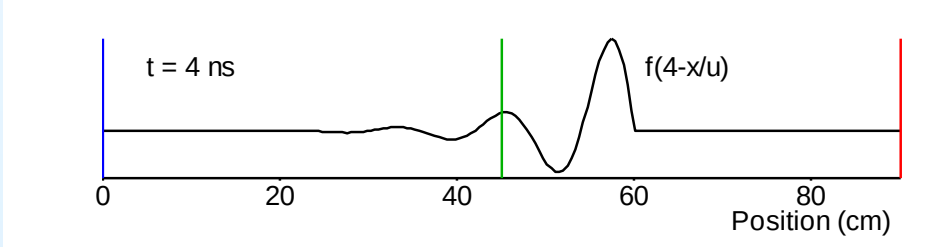

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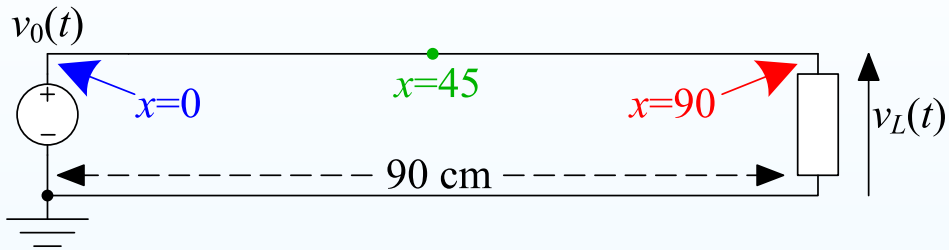

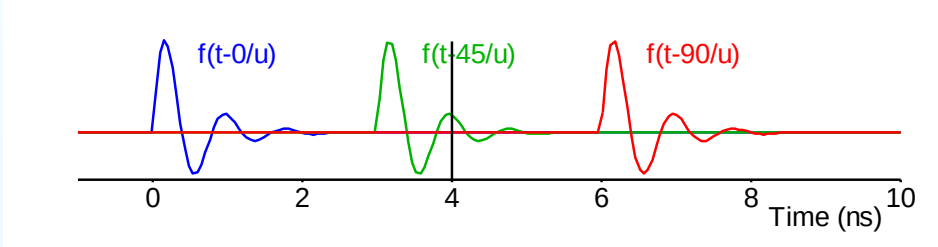

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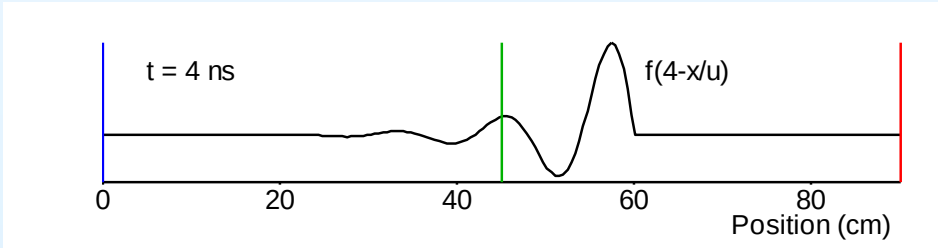

$f(t - \frac{x}{u})$ is a wave travelling forward (i.e. towards $+$x) along the line.

Forward + Backward Waves

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Similarly $g(t + \frac{x}{u})$ is a wave travelling backwards, i.e. in the $-x$ direction.

Forward + Backward Waves

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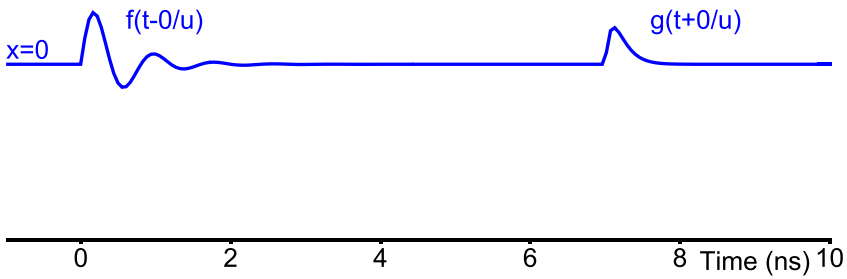
$$v(x,t) = f(t - \frac{x}{u}) + g(t + \frac{x}{u})$$



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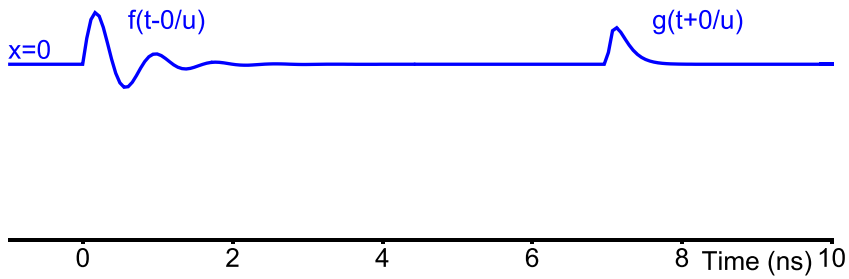

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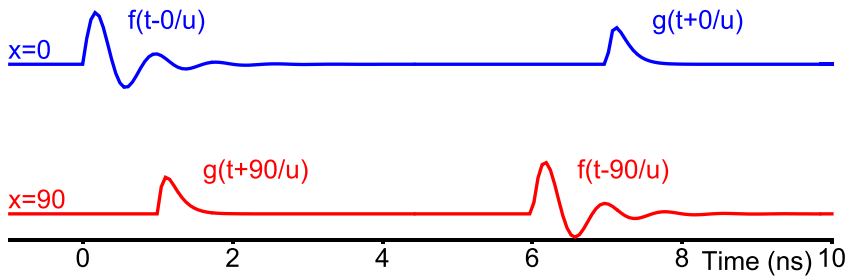

At $x = 90$ cm [▲], g starts at $t = 1$ and f starts at $t = 6$.

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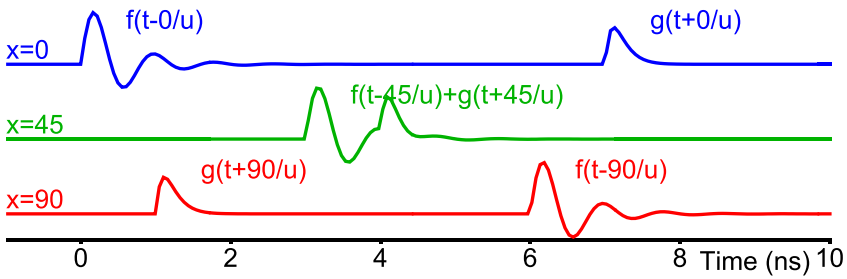

At $x = 45$ cm [▲], g is only 1 ns behind f and they add together.
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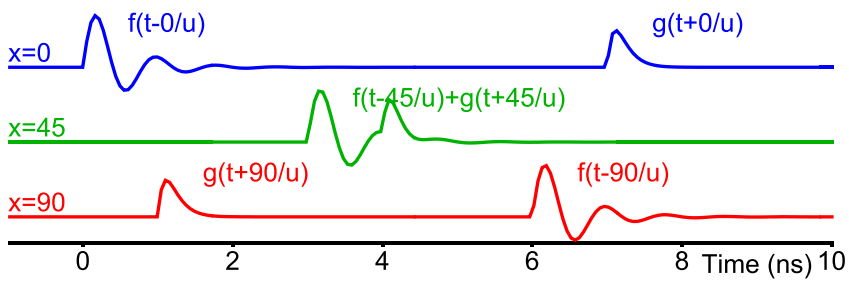

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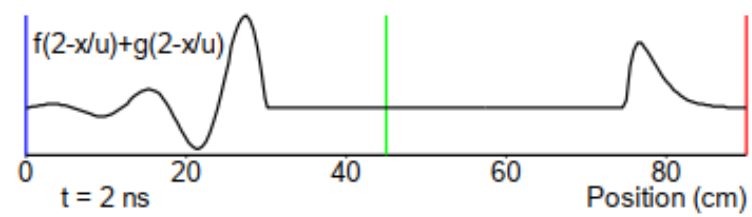
Snapshots:

At $t = 2$ ns f and g have not yet met.



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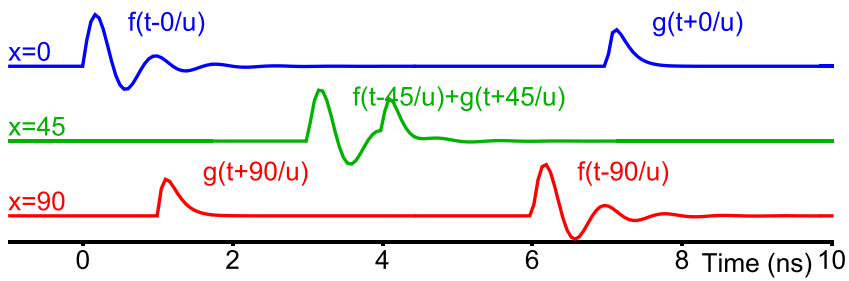

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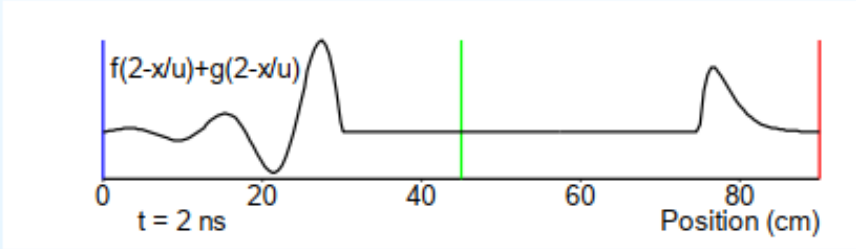
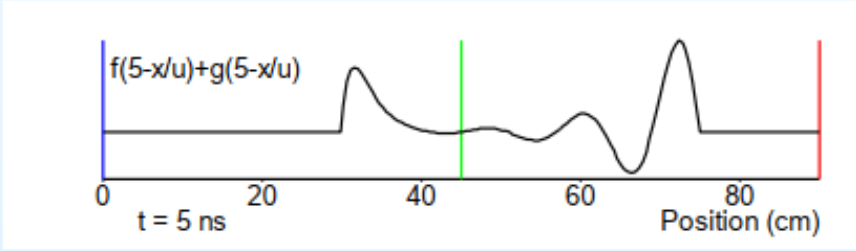


By $t = 5$ ns, f and g have just crossed.



Forward + Backward Waves

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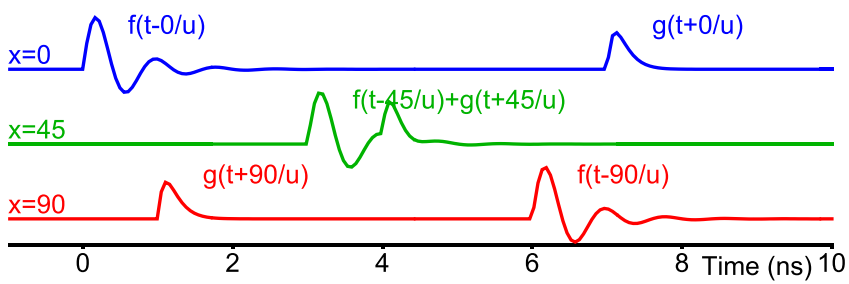

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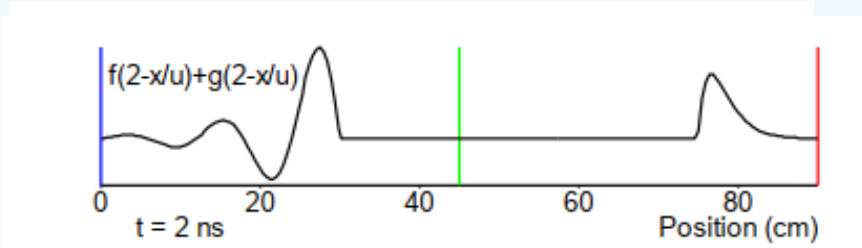

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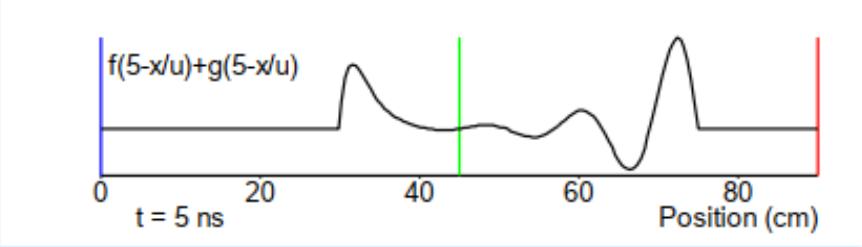

Magically, f and g pass through each other entirely unaltered

Power Flow

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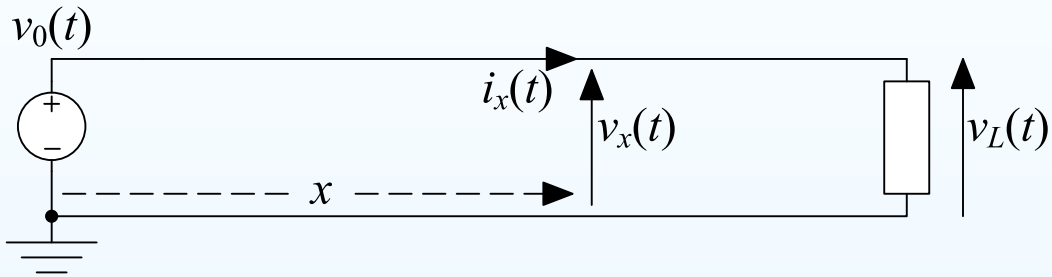
Define $f_x(t) = f\left(t - \frac{x}{u}\right)$ and $g_x(t) = g\left(t + \frac{x}{u}\right)$ to be the forward and backward waveforms at any point, x.



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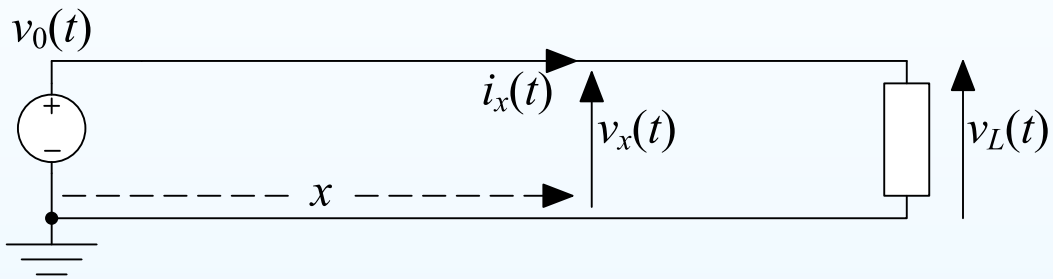

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Then $\quad v_x(t) = f_x(t) + g_x(t) \quad$ and $\quad i_x(t) = Z_0^{-1}\left(f_x(t) - g_x(t)\right).$

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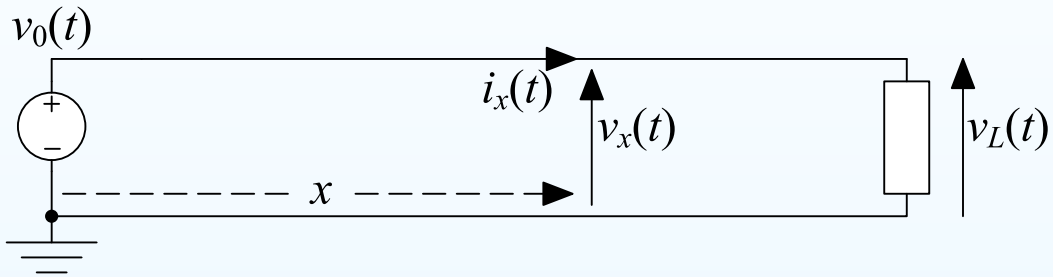

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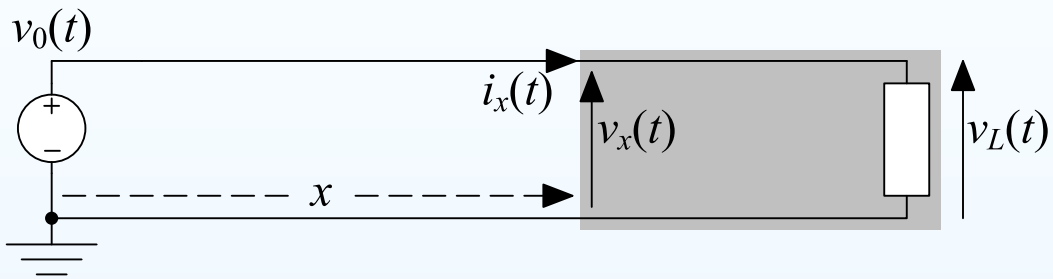

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$$P_x(t) = v_x(t) i_x(t)$$

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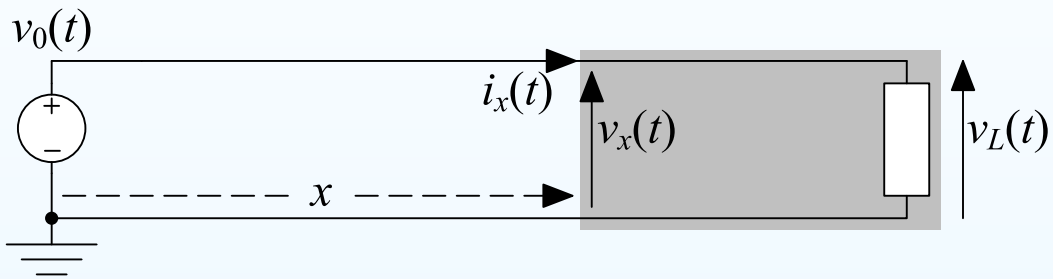

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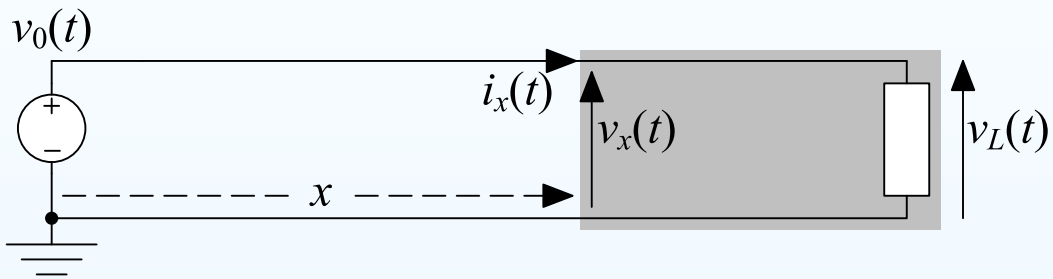

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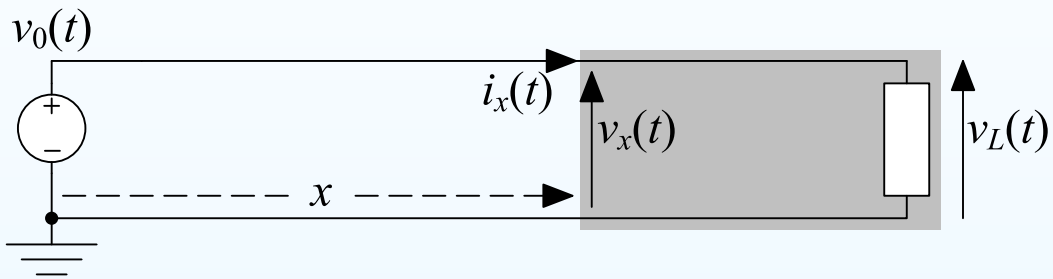

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f_x carries power into shaded area and g_x carries power out independently.

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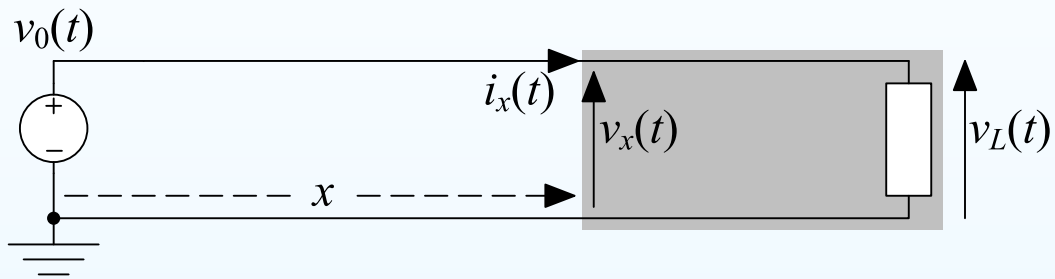

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f_x carries power into shaded area and g_x carries power out independently.

Power travels in the same direction as the wave.

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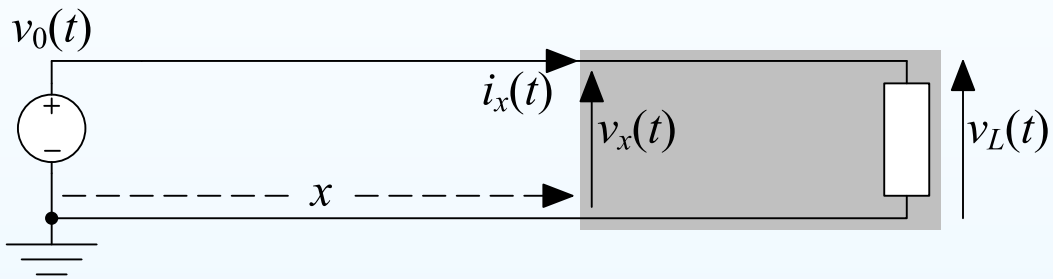

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The same power as would be absorbed by a [ficticious] resistor of value Z_0.

Reflections

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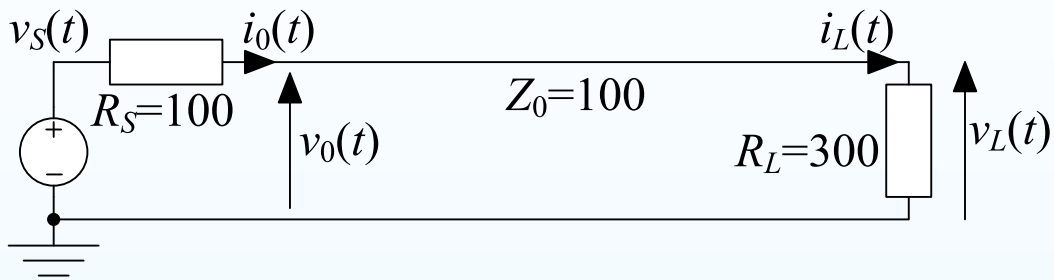

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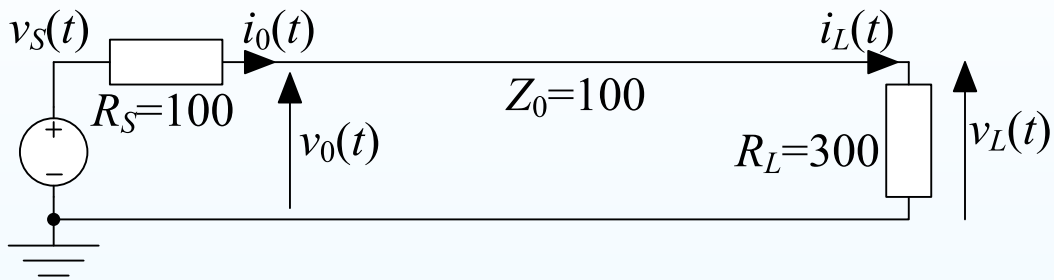

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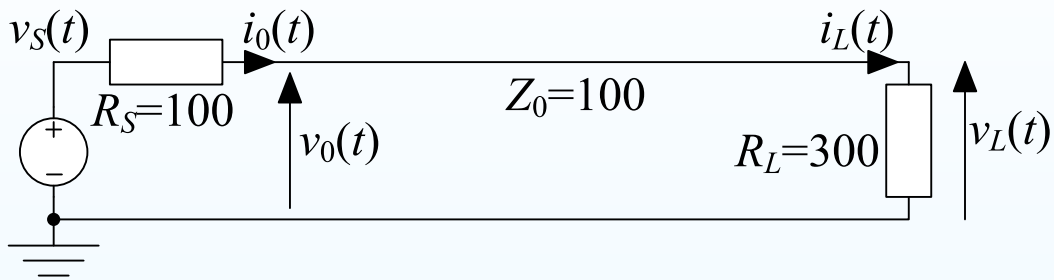

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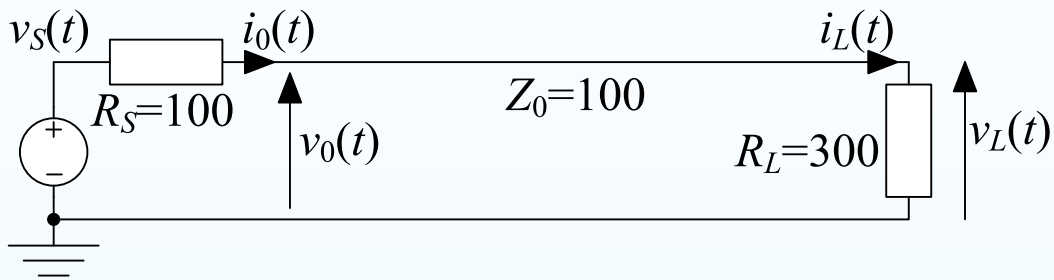

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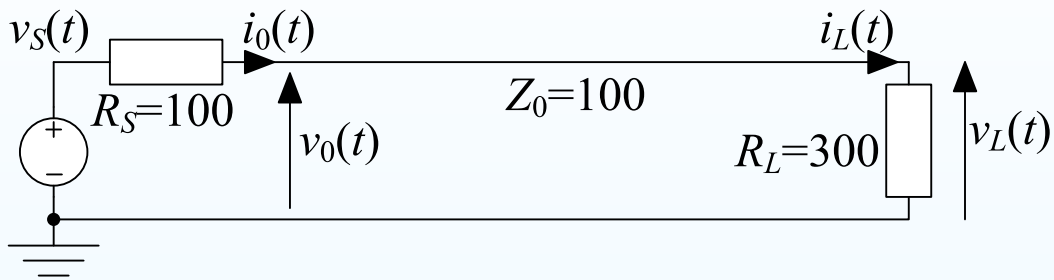

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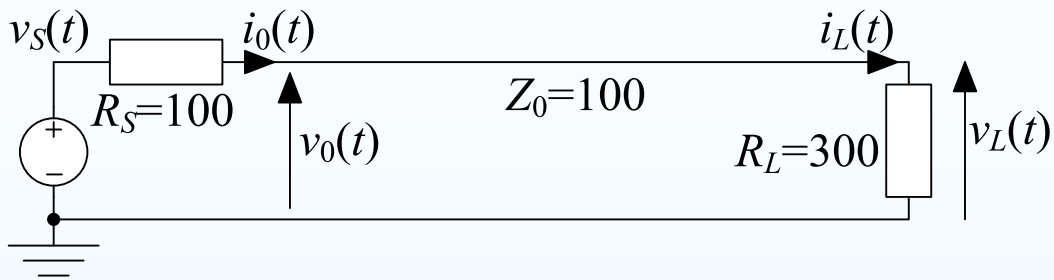

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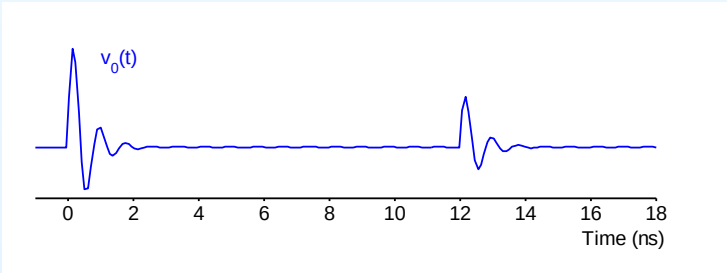

At source end: $g_0(t) = \rho_L f_0\left(t - \frac{2L}{u}\right)$ i.e. delayed by $\frac{2L}{u} = 12$ ns.

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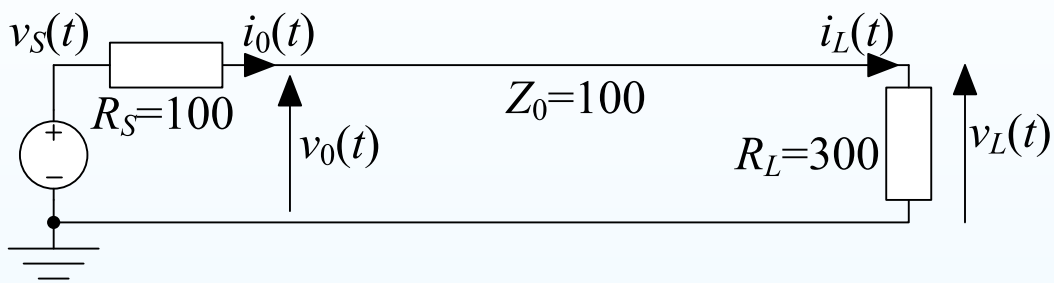

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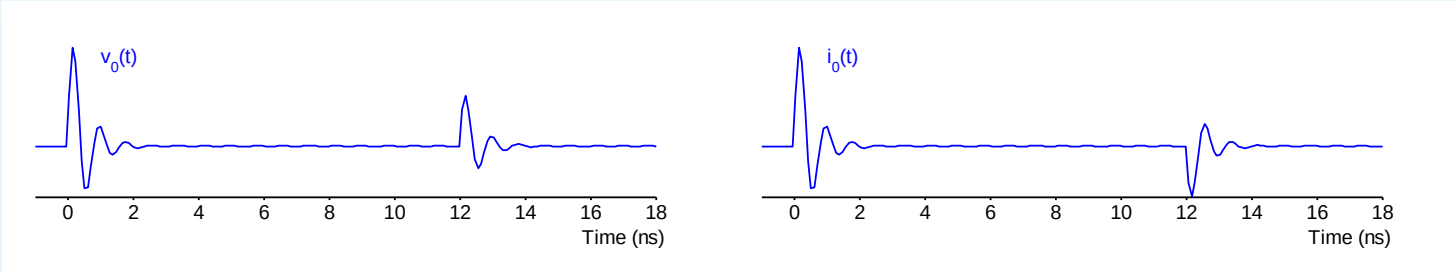

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Note that the reflected current has been multiplied by $-\rho$.

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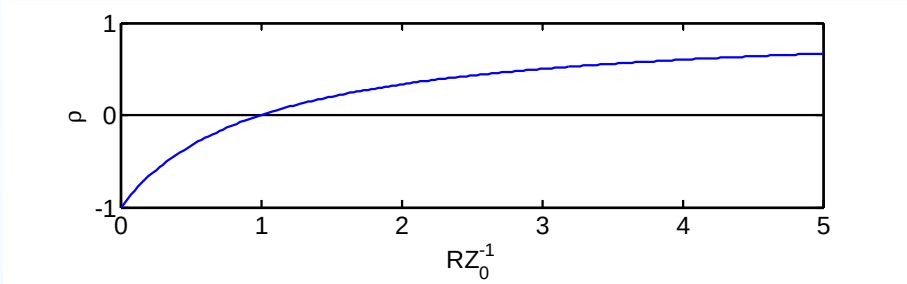
Reflection Coefficients

$$\rho = \frac{R - Z_0}{R + Z_0} = \frac{\frac{R}{Z_0} - 1}{\frac{R}{Z_0} + 1}$$



ρ depends on the ratio $\frac{R}{Z_0}$.

$\frac{R}{Z_0}$	ρ	$\frac{v_L(t)}{f(t)}$	$\frac{i_L(t)Z_0}{f(t)}$	Comment
3	$+0.5$			

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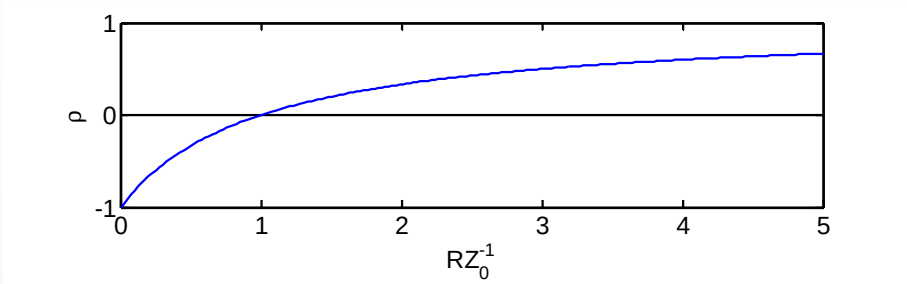

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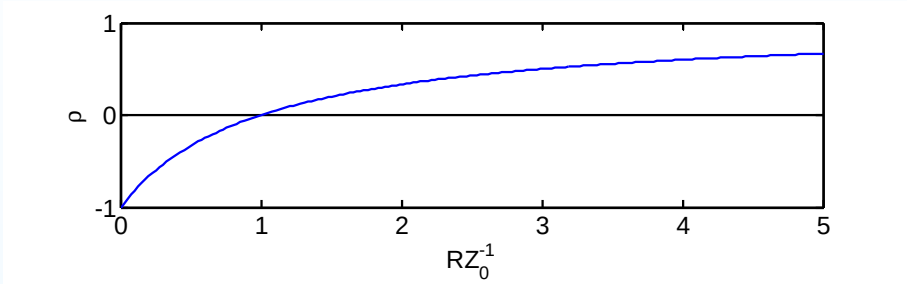

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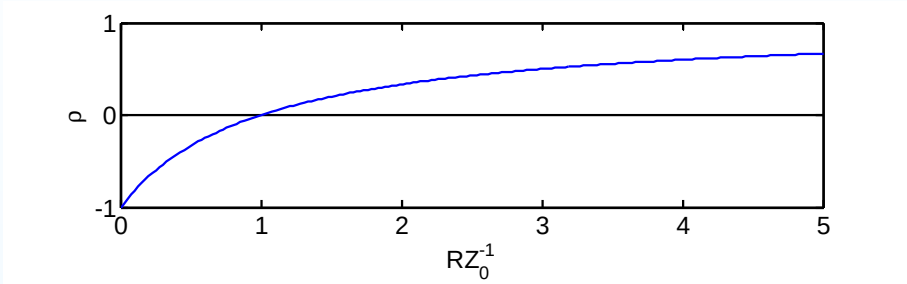

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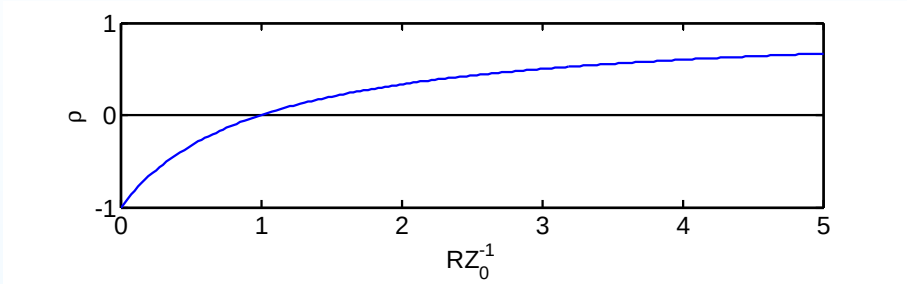

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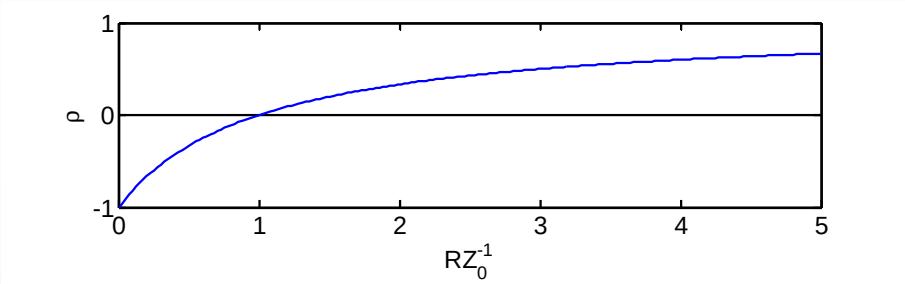

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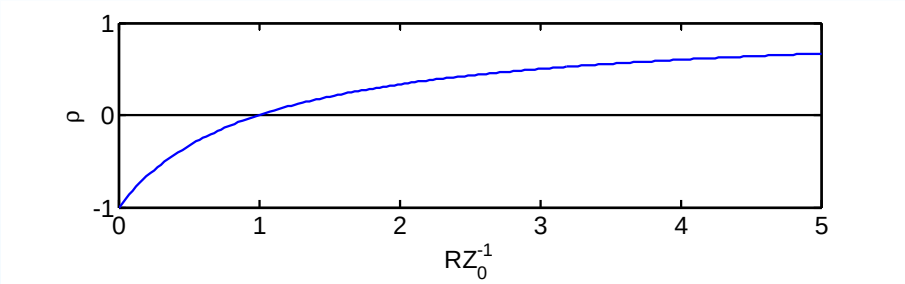

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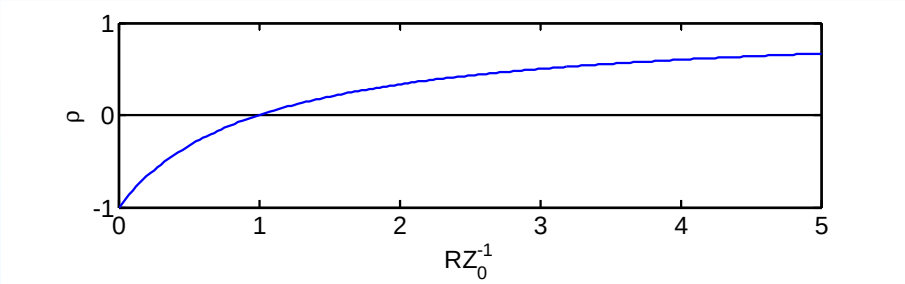

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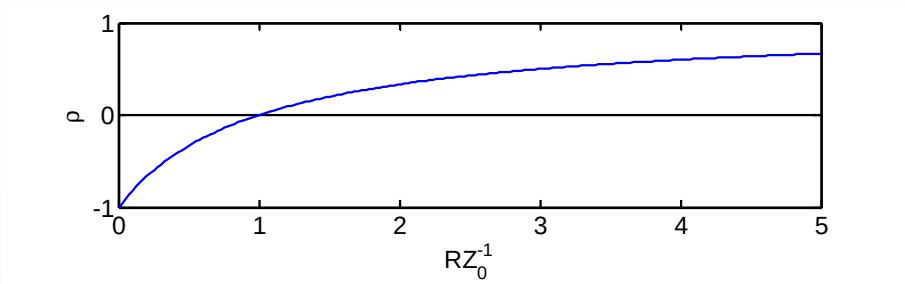

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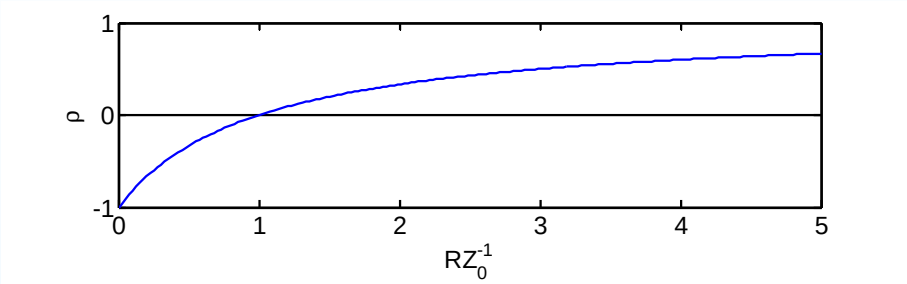

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Remember: $\rho \in \{-1, +1\}$ and increases with R.

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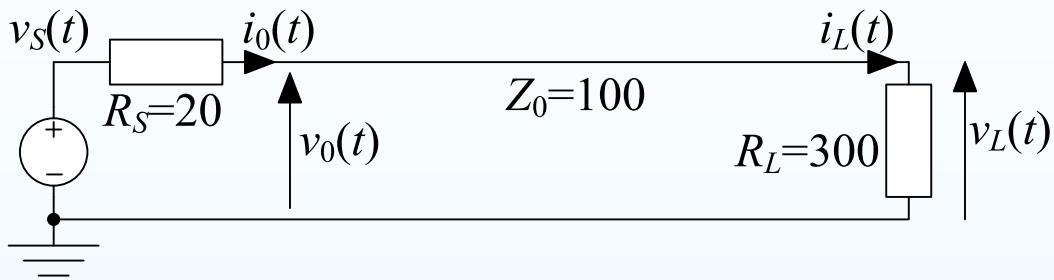

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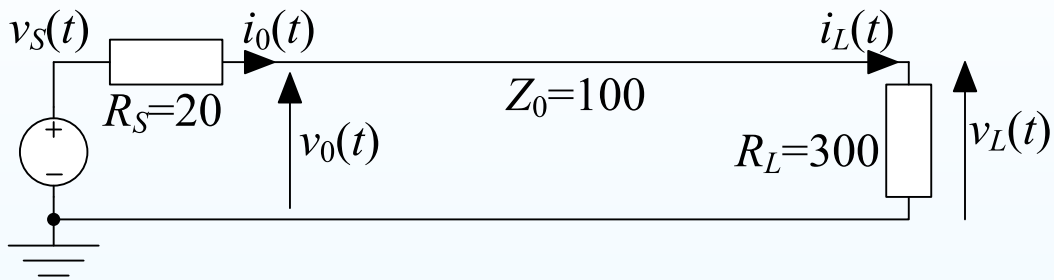

$$v_x = f_x + g_x$$
$$i_x = \frac{f_x - g_x}{Z_0}$$

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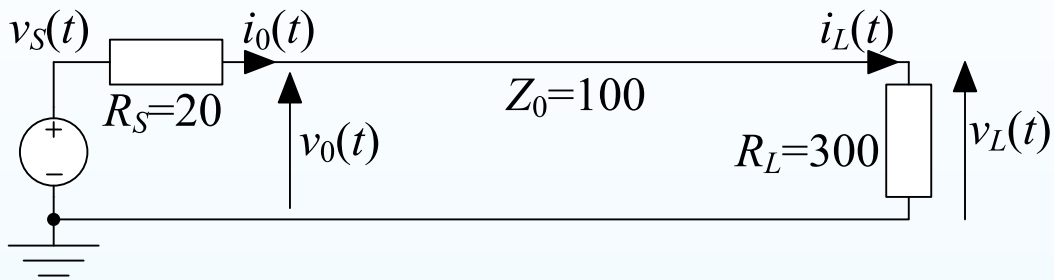

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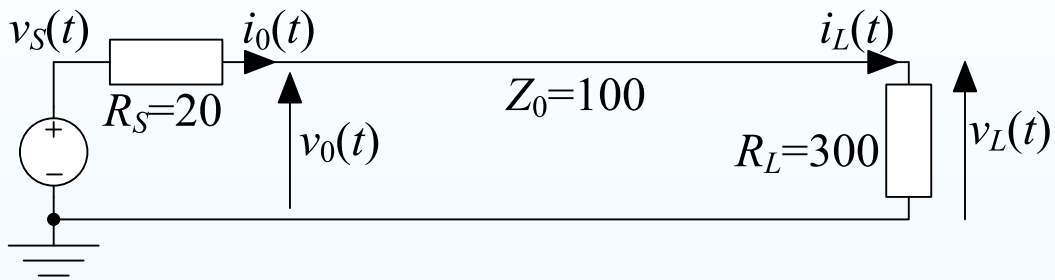

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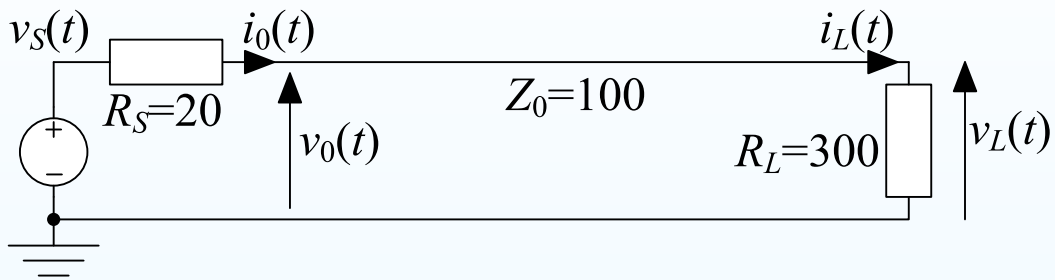

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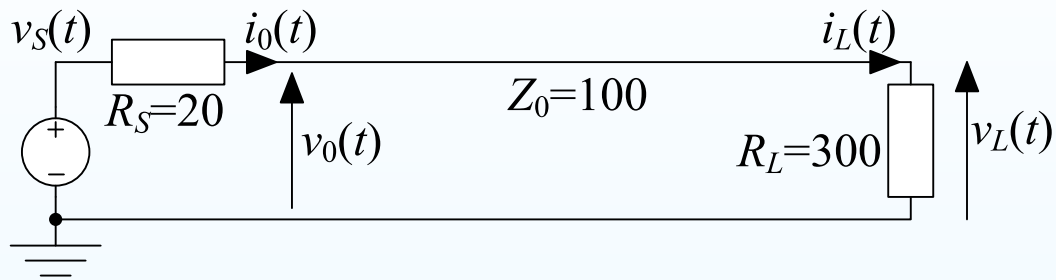

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For $R_S = 20$: $\tau_0 = \frac{100}{20+100} = 0.83$ and $\rho_0 = \frac{20-100}{20+100} = -0.67$.

Multiple Reflections

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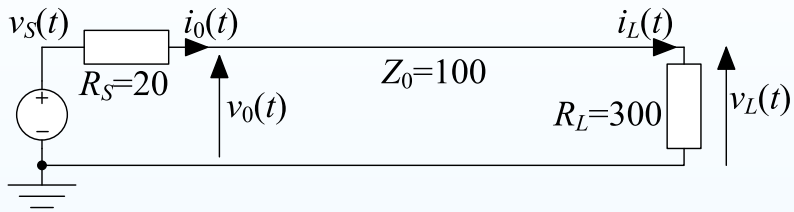

$$\rho_0 = -\frac{2}{3}$$
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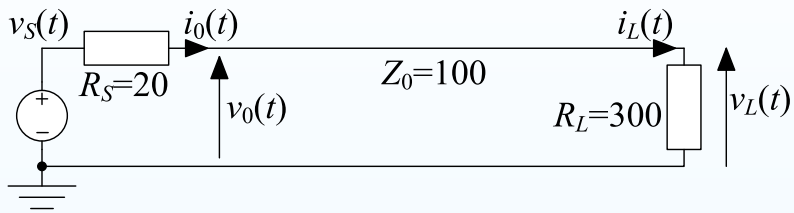

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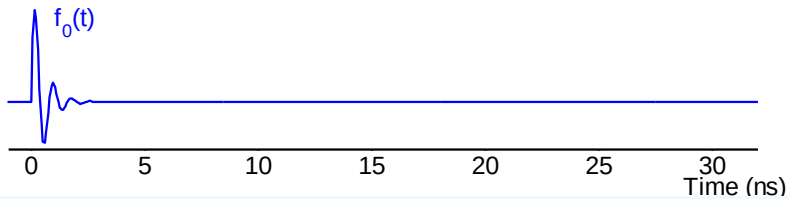

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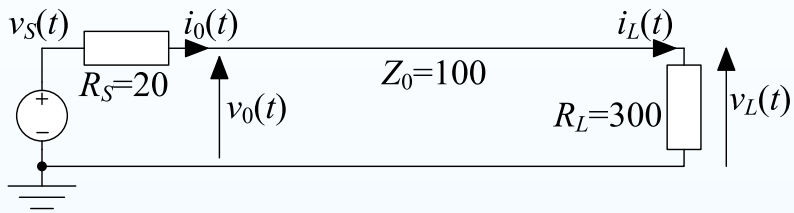

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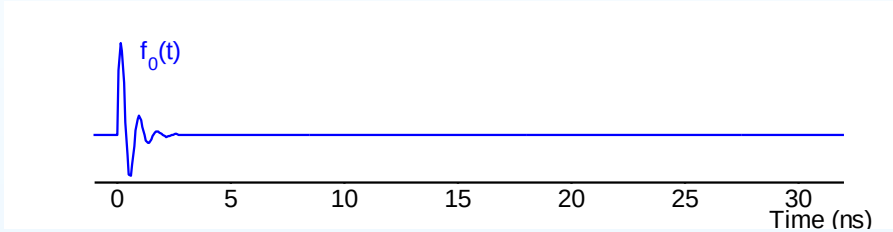

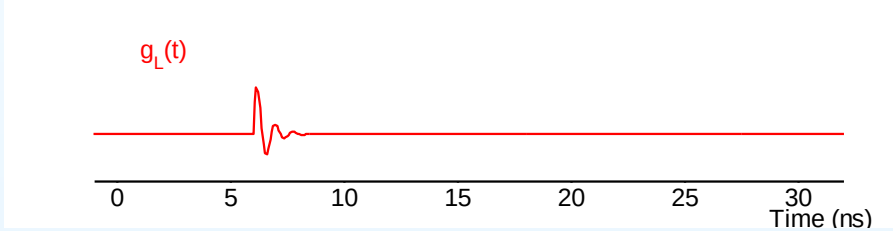

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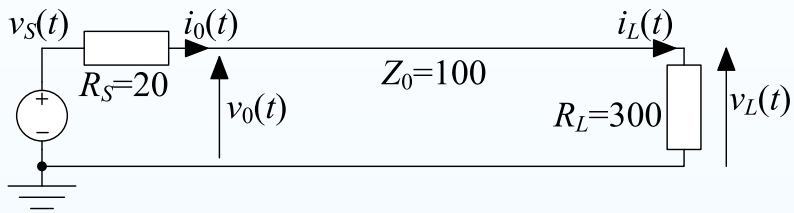

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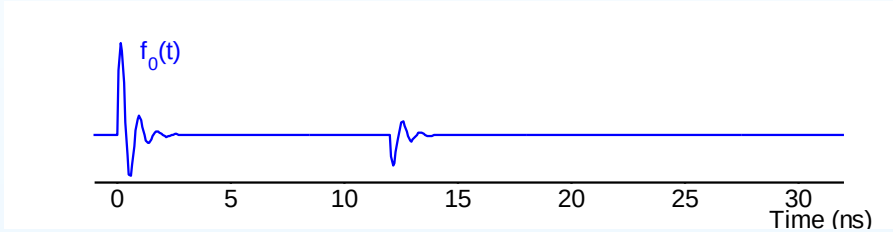

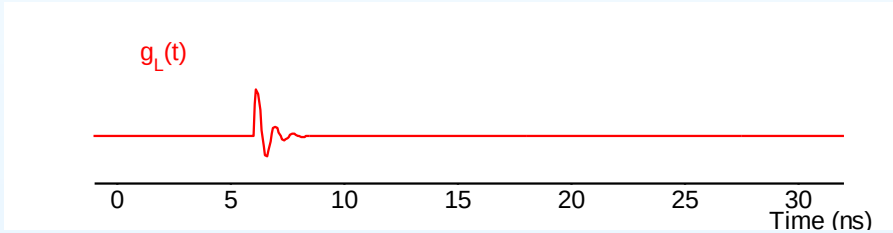

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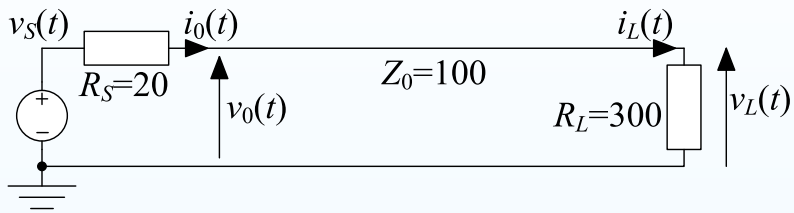

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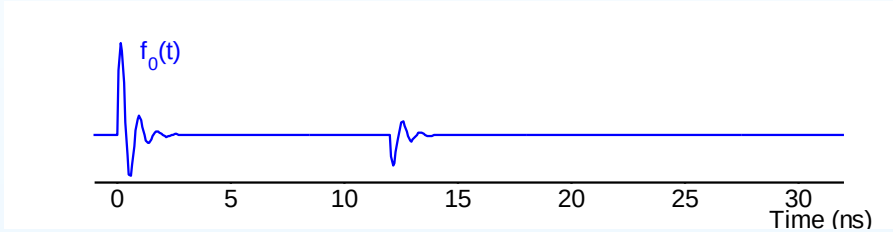

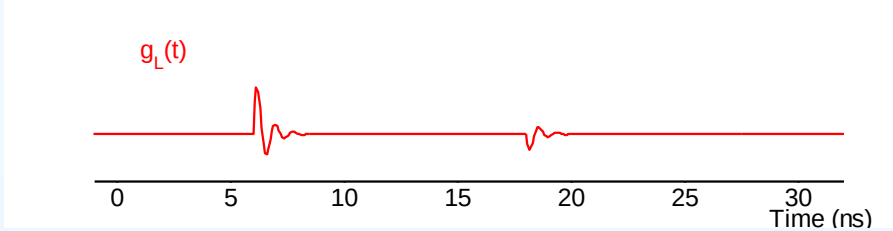

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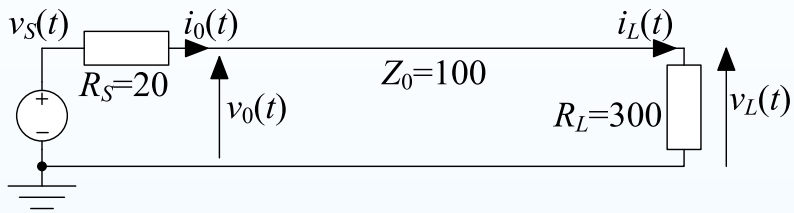

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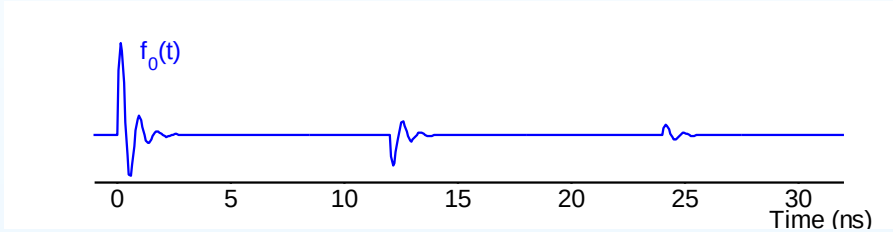

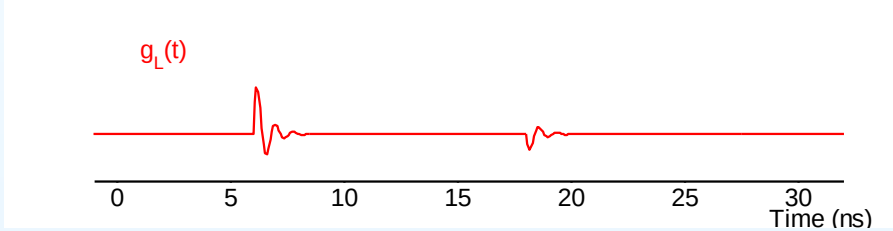

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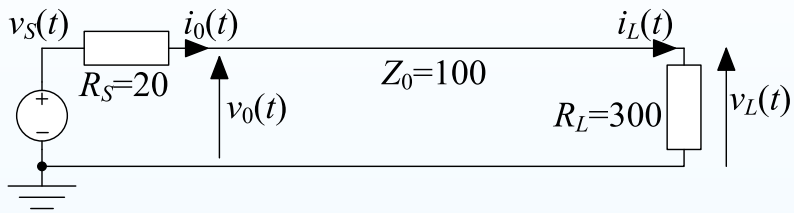

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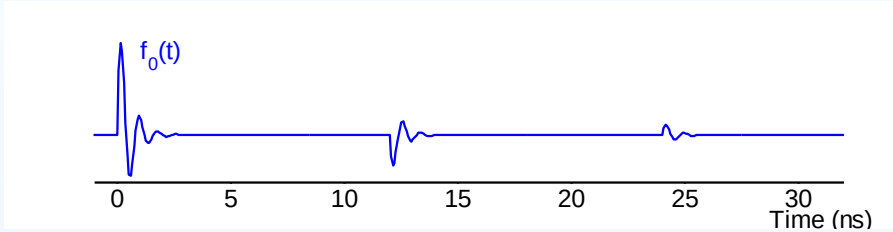

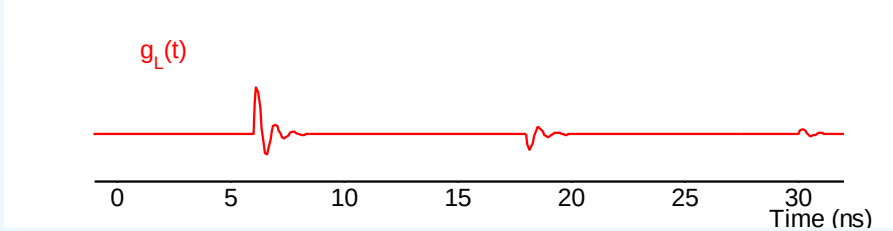

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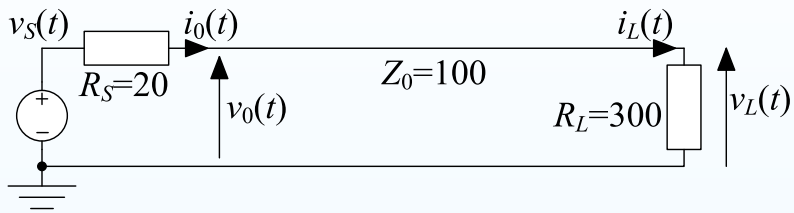

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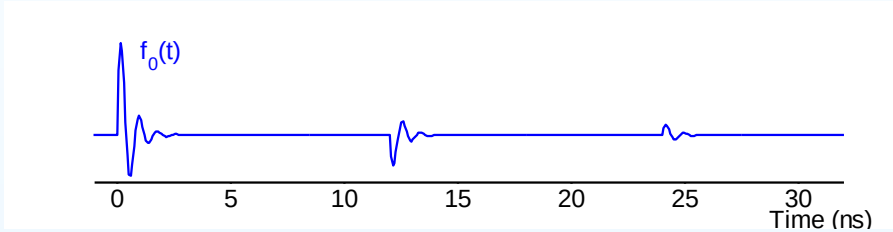

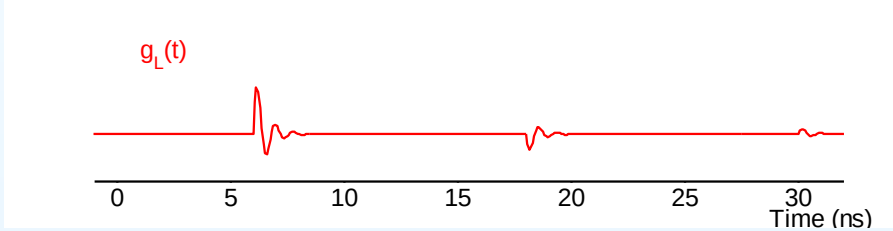

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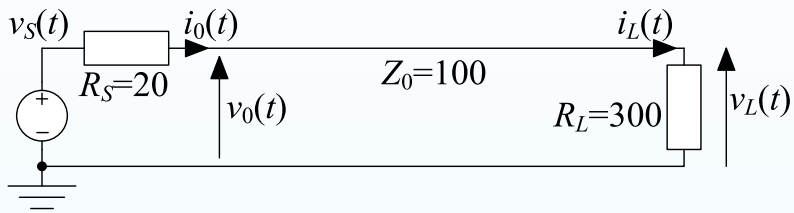

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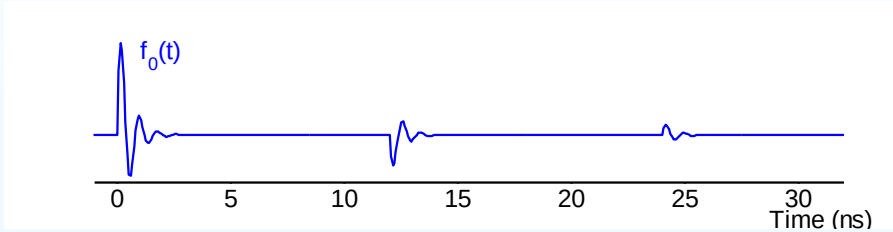

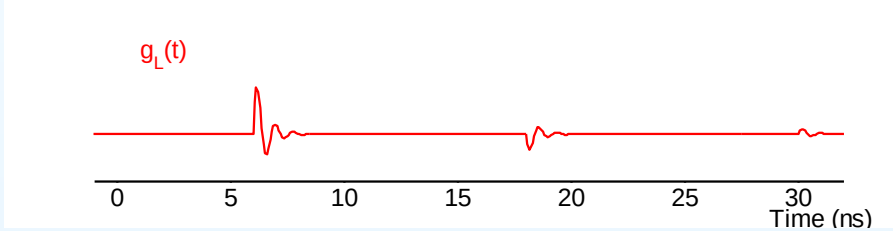

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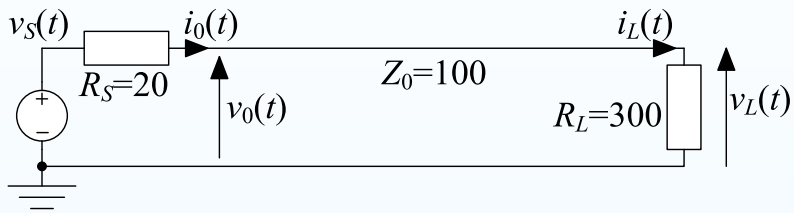

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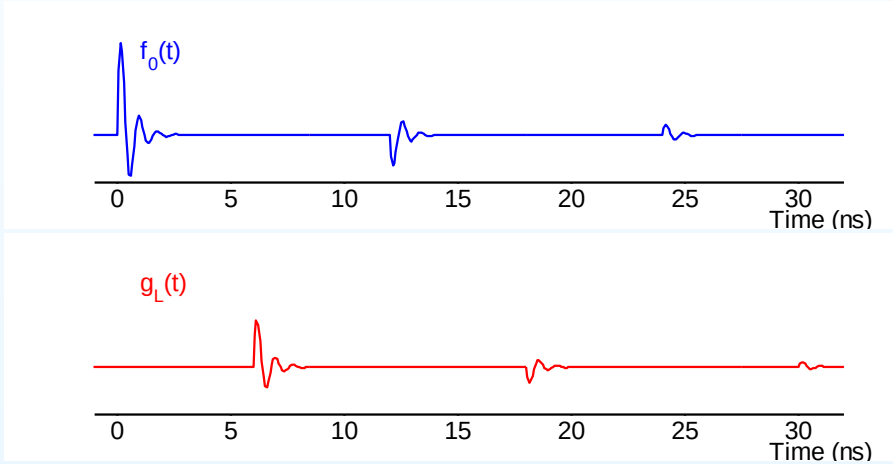

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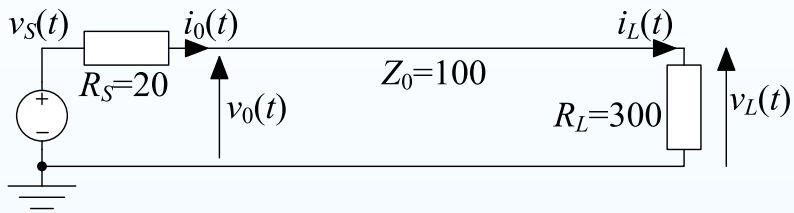

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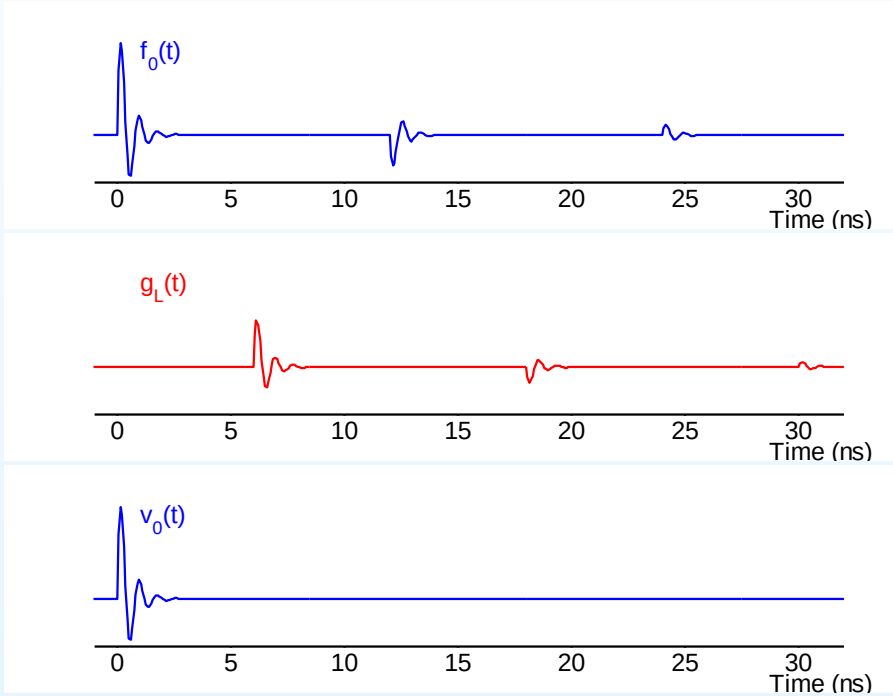

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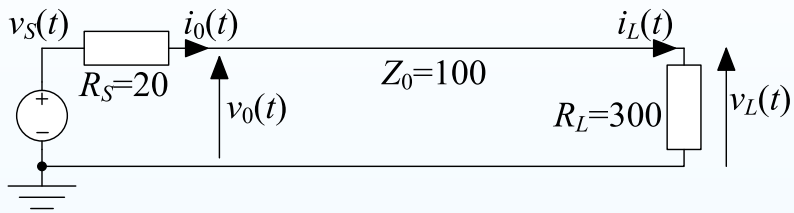

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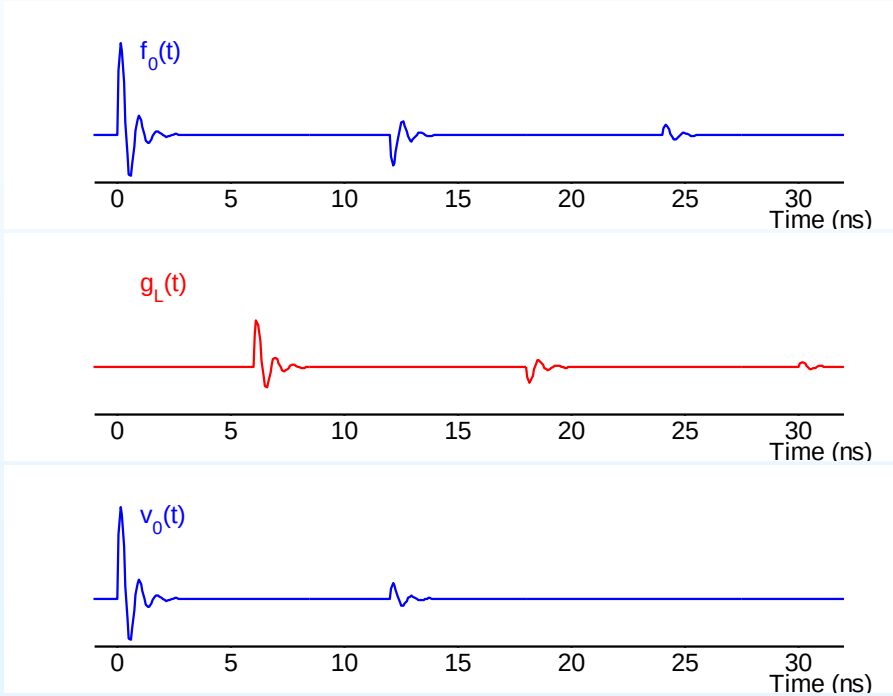

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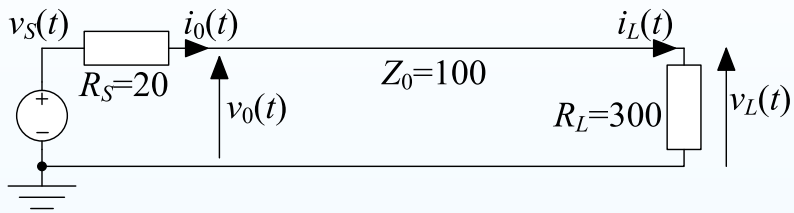

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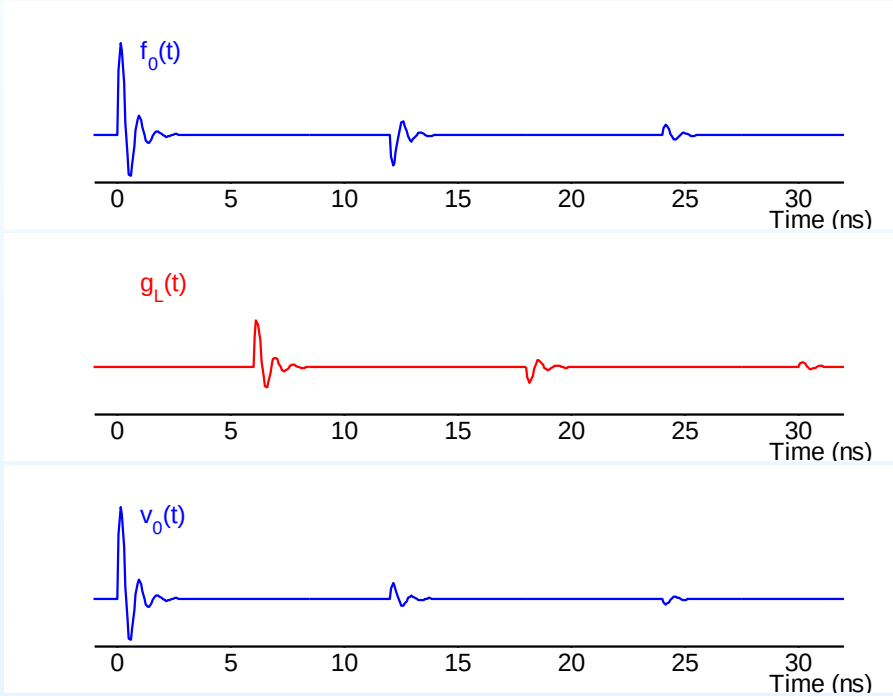

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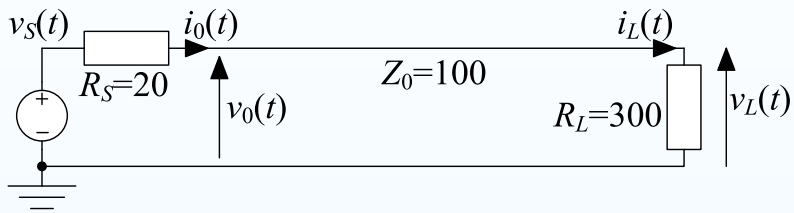

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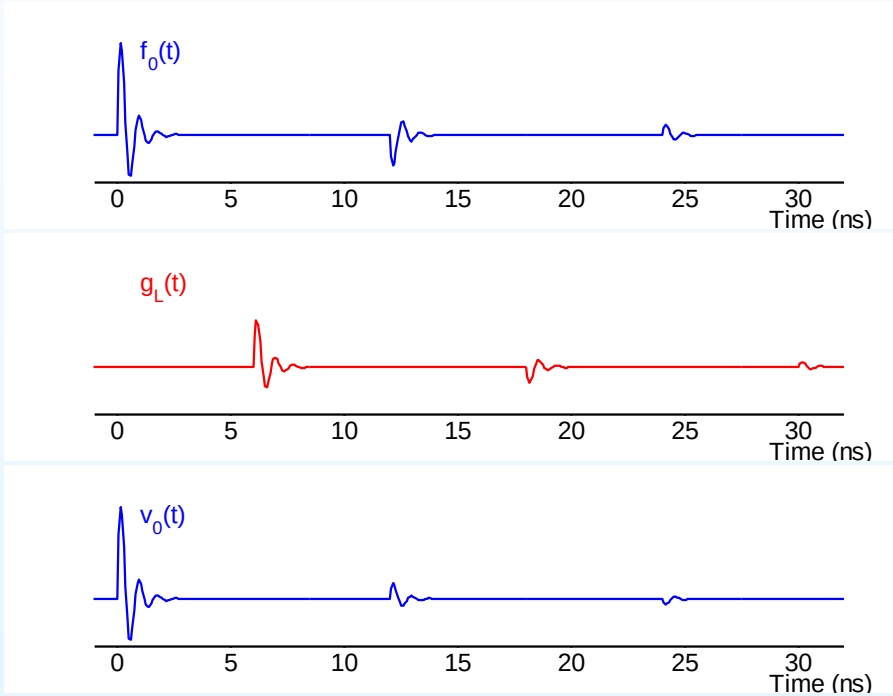

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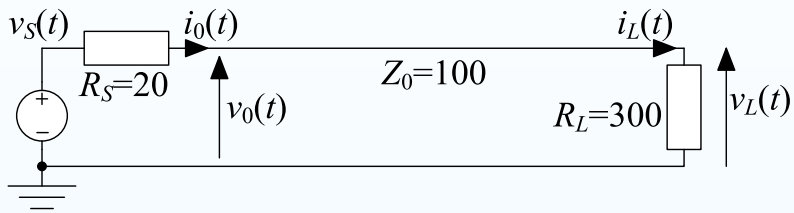

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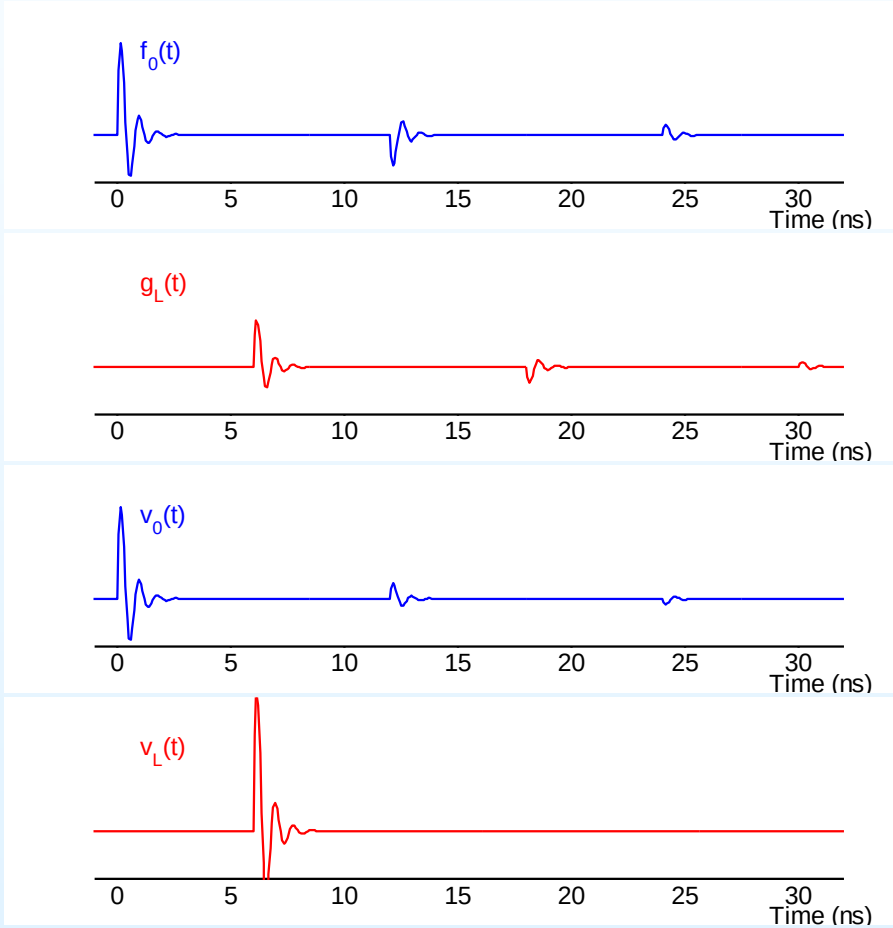

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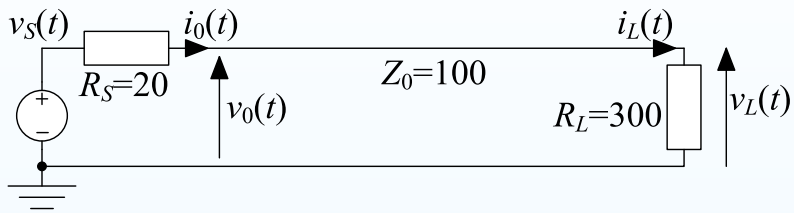

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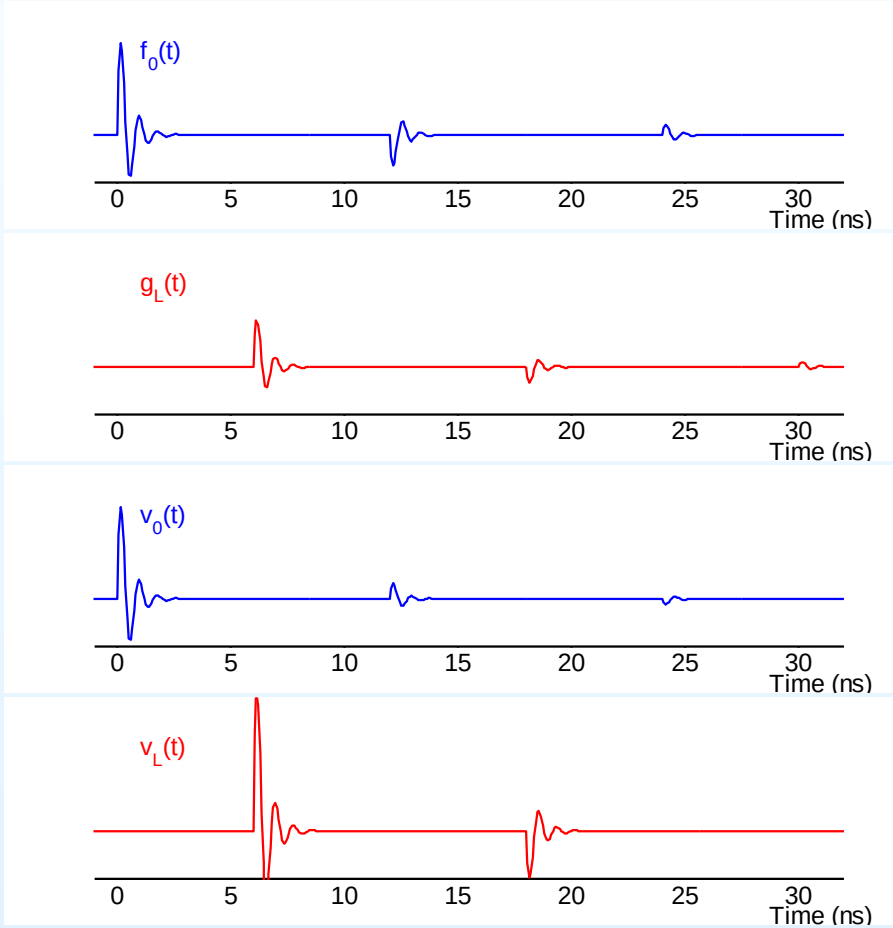
$$g_L(t) = \rho_L f_0\left(t - \frac{L}{u}\right)$$

$$v_0(t) = f_0(t) + g_L\left(t - \frac{L}{u}\right)$$

$$v_L(t) = f_0\left(t - \frac{L}{u}\right) + g_L(t)$$



Multiple Reflections

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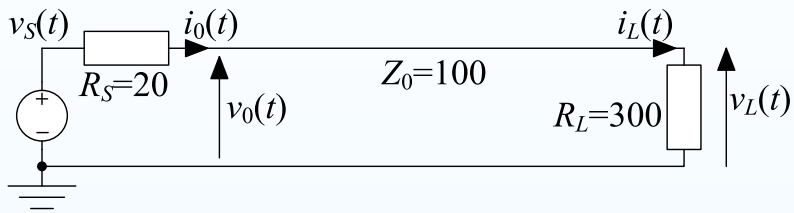

$$\rho_0 = -\frac{2}{3}$$
$$\rho_L = \frac{1}{2}$$
$$v_x = f_x + g_x$$

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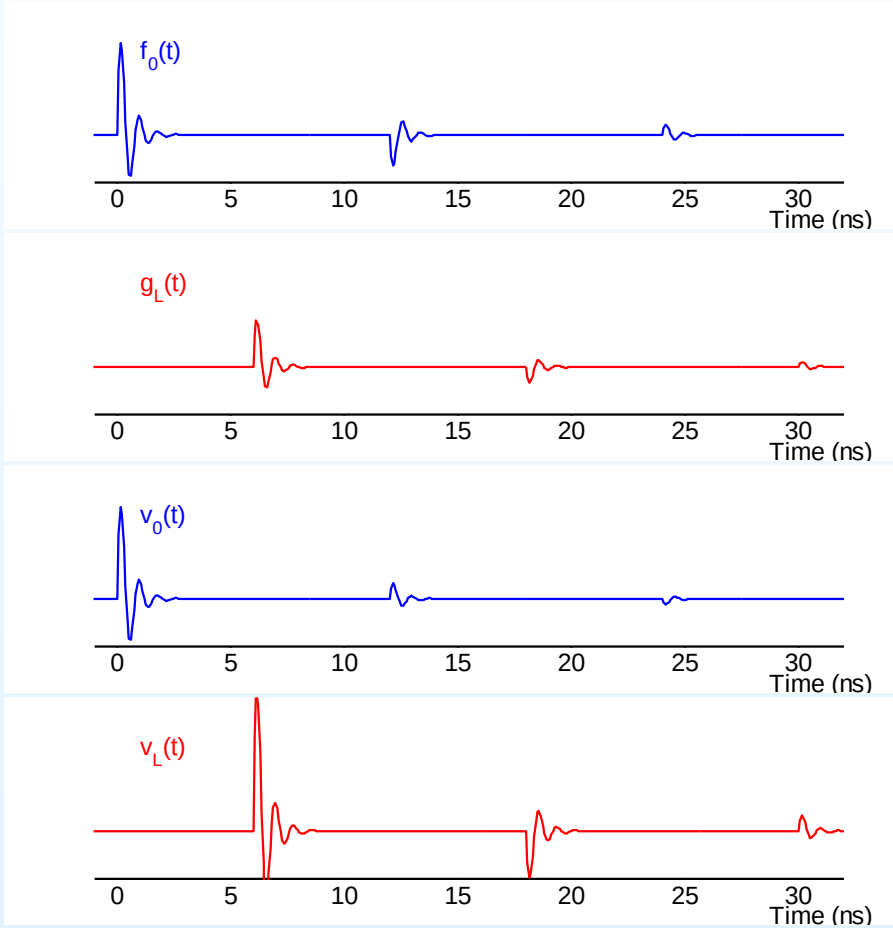

Transmission Line Characteristics

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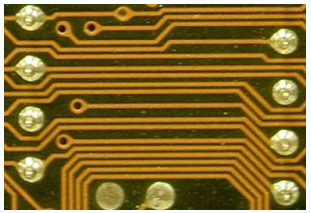
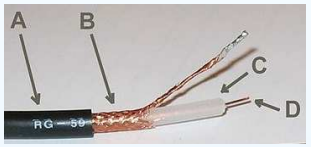
Integrated circuits & Printed circuit boards



High speed digital or high frequency analog interconnections

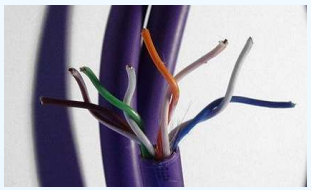
$Z_0 \approx 100\,\Omega$, $u \approx 15$ cm/ns.

Long Cables



Coaxial cable ("coax"): unaffacted by external fields; use for antennae and instrumentation.

$Z_0 = 50$ or $75\,\Omega$, $u \approx 25$ cm/ns.



Twisted Pairs: cheaper and thinner than coax and resistant to magnetic fields; use for computer network and telephone cabling. $Z_0 \approx 100\,\Omega$, $u \approx 19$ cm/ns.

When do you have to bother?

Answer: long cables or high frequencies. You can completely ignore transmission line effects if $\text{length} \ll \frac{u}{\text{frequency}} = \text{wavelength}$.

- Audio (< 20 kHz) never matters.
- Computers (1 GHz) usually matters.
- Radio/TV usually matters.

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  Only matters for high frequencies or long cables.

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$$f_x(t) = f_0\left(t - \frac{x}{u}\right) \quad \text{and} \quad g_x(t) = g_0\left(t + \frac{x}{u}\right)$$

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  - $R = Z_0$ avoids reflections: *matched* termination.
  - Reflections go on for ever unless one or both ends are matched.
  - f is infinite sum of copies of the input signal delayed successively by the round-trip delay, $\frac{2L}{u}$, and multiplied by $\rho_L \rho_0$.